



TEXT CLASSIFICATION IN NLP: MEDIA BIAS
DETECTION, SEXISM DETECTION, AND [STUDENT
3 TOPIC]

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Abstract

In this paper we will explore text classification on three different NLP applications; media bias detection on news articles, sexism detection on social media comments, and [Student 3 task]. The three tasks, taken together, illustrate a range of the types of classification found within NLP. Three deep learning architectures will be experimented on for each task; a Bidirectional LSTM trained from scratch, the two pre-trained transformers BERT, and RoBERTa. The models are fine-tuned on task specific data and measured by accuracy and F1-score. For media bias detection, RoBERTa outperformed all models yielding an accuracy of 84.60%, and a weighted F1-score of 0.8464. For sexism detection, BERT performed best with an accuracy of 85.65% and a macro F1-score of 0.8115. The three tasks demonstrate a trend across the models; transformers outperforms the BiLSTM.

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Introduction

Text classification is one of the foundational tasks in NLP and underpins many practical applications we use on a daily basis. At its most basic, text classification assigns a label from a predetermined set to an instance of text based on its contents, providing a pathway to automating understanding over large scales. These applications include, but are not limited to: detecting media bias and misinformation in news [7]; classifying the argumentative intent of scientific documents [10]; and identifying harmful content in social media [11].

While the formulation of text classification may be seemingly simple (the assignment of a discrete label to an item of text), the complexities of human language provide a variety of challenges to be addressed in text classification such as learning the semantics and contextuality of words. Early techniques relied on manually engineered features, shallow machine learning models such as Naive Bayes [1] and Support Vector Machines [2], and more recently recurrent neural networks, particularly Long Short-Term Memory networks [3], began learning representations directly from the text. Most impactfully however, was the introduction of the Transformer [4] and subsequently, the pre-training and subsequent release of very large scale language models such as BERT [5] and RoBERTa [6].

This project looks at text classification in three different NLP tasks each corresponding to a different area of application. The group then proceeds to individually train three different models (BiLSTM, BERT, RoBERTa) on three individually chosen datasets and reports their performance in terms of accuracy and weighted F1-score. The intention of this is to compare how the three models behave under different circumstances.

Chapter 1

Atakan Gül — Media Bias Detection in News

1.1 Methodology

This work targets the task of media bias detection from news articles. It is a classification task where the model has to decide whether a news sentence has biased framing or no bias. Media bias detection is a non-trivial NLP problem because bias is manifested by subtler things like phrasing, framing and tone, and not by stating false facts [7].

In this work, 3 models are tested:

Bidirectional LSTM. A Long Short-Term Memory network is a type of recurrent neural network designed to remember information for long periods of time [3]. Bidirectional LSTMs are trained in both directions in order to make better use of the context. In our work the model takes an embedding layer, two layers of 256-hidden-dimension Bidirectional LSTMs and a linear classification layer on top of the concatenation of the last forward and backward hidden states.

BERT. BERT is a pre-trained language model based on the Transformer encoder architecture. It is trained on a large corpus using a masked language modeling objective and a next sentence prediction objective and therefore is able to produce deep bidirectional context representations [5]. We feed the [CLS] token output through a linear classifier. For the implementation of BERT we use the `bert-base-uncased` variant.

RoBERTa. RoBERTa is a BERT variant that removes next sentence pre-

diction, uses dynamic masking, and is trained with larger corpora and batch sizes [6]. This architecture has been shown to perform consistently better on downstream tasks. For RoBERTa we use the **roberta-base** variant.

The transformer models (BERT and RoBERTa) are fine-tuned from the beginning with an AdamW optimizer and a linear learning rate scheduler with warmup. The LSTM model is trained from scratch using the Adam optimizer.

1.2 Experiments & Findings

Dataset. The BABE dataset (Bias Annotations By Experts), introduced by Spinde et al. [8], is employed for this task. The Hugging Face distribution used in this project contains 4,121 news sentences gathered from a diverse collection of sources, each rated as either biased (1) or not biased (0) by trained annotators; it is divided into 3,121 training and 1,000 test samples [9]. Since the labeling is conducted with the help of expert annotators and has been designed carefully, it serves as a reliable reference for media bias detection.

Training Configuration. For all models, up to 3 epochs of training are performed with early stopping monitored on a held-out validation set (10% of the training data), using a minimum improvement threshold of 0.5% weighted F1 and a patience of 2 epochs. A batch size of 16 and a maximum sequence length of 128 tokens are used. The learning rate is 2×10^{-5} for transformer models and 1×10^{-3} for the LSTM. All experiments are run on an NVIDIA GeForce RTX 3060 Laptop GPU.

Results. Table 1.1 presents the performance of each model on the BABE test set.

Model	Accuracy	Weighted F1
LSTM	0.6560	0.6511
BERT	0.8260	0.8266
RoBERTa	0.8460	0.8464

Table 1.1: Model performance on the BABE dataset for media bias detection.

Figure 1.1 provides a visual comparison of accuracy and F1 scores across all three models. Figure 1.2 shows the training loss per epoch for each model. Figure 1.3 presents the confusion matrices on the test set.

Qualitative Examples. Table 1.2 presents four representative test sentences selected to illustrate three patterns: cases where all models agree correctly, cases where BiLSTM fails but both transformers succeed, and a case where BERT and RoBERTa disagree.

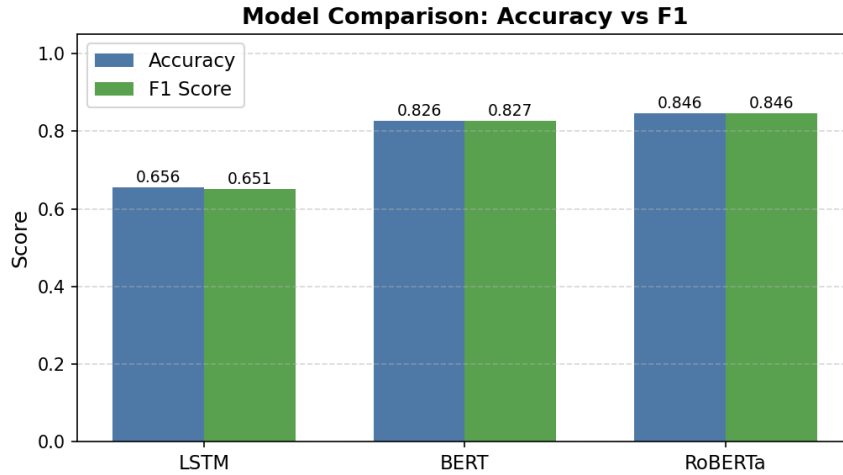


Figure 1.1: RoBERTa achieves the highest accuracy and weighted F1, with both transformer models substantially outperforming BiLSTM.

1.3 Discussion

It’s noticeable from these numbers that there is a hierarchy of performance among the three architectures. The Bidirectional LSTM attained 65.60% accuracy and weighted F1 of 0.6511, which is better than random chance, but quite poor relative to the other architectures based on transformers. These numbers also suggest the challenge of learning both representations and task-specific patterns without any pretraining on the relatively small media bias dataset.

BERT has promising results, reaching 82.60% accuracy and a weighted F1 of 0.8266, which highlights the advantages of transfer learning. As can be seen from the confusion matrix in Figure 1.3, BERT was able to classify both classes more consistently, with far fewer incorrect classifications of biased sentences compared to the LSTM.

The best performing model in this experiment is RoBERTa, with an accuracy of 84.60% and a weighted F1 of 0.8464. The authors of RoBERTa attributed the improvement over BERT to a set of changes made to the BERT pre-training objective: discarding the next sentence prediction task, implementing dynamic masking of input tokens, using larger batches and more training data [6]. The impact of this change can be seen in the gains over BERT: 2.00 accuracy percentage points (84.60% vs 82.60%) and 0.0198 weighted F1 (0.8464 vs 0.8266) on the media bias detection task.

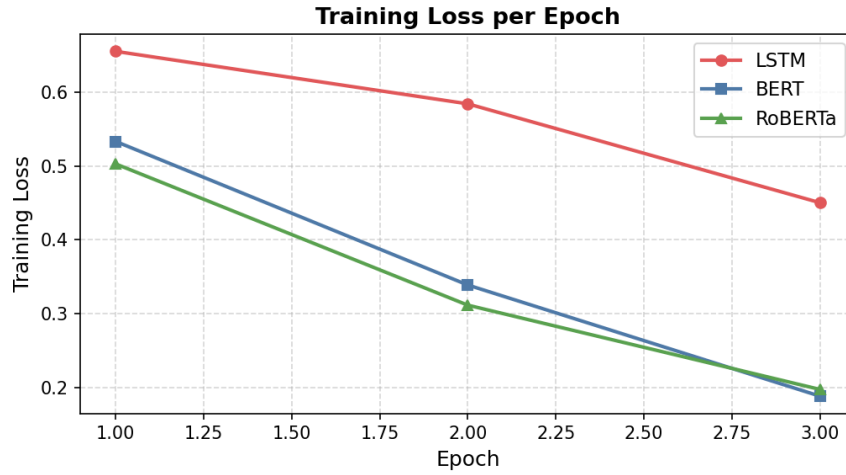


Figure 1.2: Training loss decreases across all three models; transformer models start lower and converge faster than BiLSTM.

1.4 Conclusion

We built three different architectures of deep learning models for media bias detection on BABE dataset. The experimental results demonstrate that pre-trained transformers models obtained substantially superior performance to the LSTM baseline, and RoBERTa obtained the best performance: 84.60

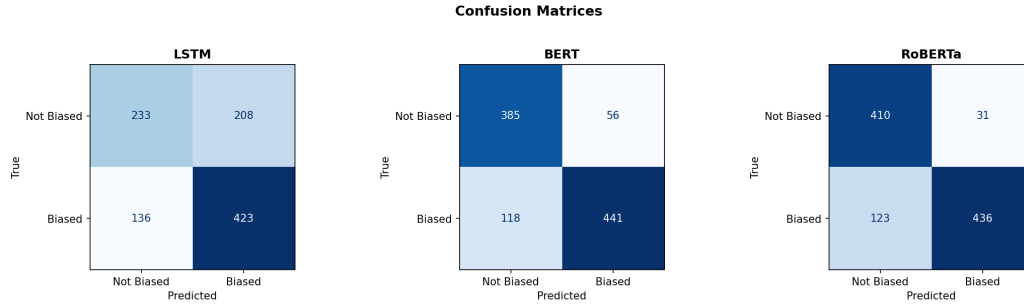


Figure 1.3: Confusion matrices showing RoBERTa and BERT classify both classes more reliably than BiLSTM, which struggles particularly on biased sentences.

Sentence	True	LSTM	BERT	RoBERTa
<i>‘Nearly 78% of Americans report experiencing concern about the climate...’</i>	NB	NB ✓	NB ✓	NB ✓
<i>Companies like Nike and Google have made Juneteenth a paid holiday.”</i>	NB	B ×	NB ✓	NB ✓
<i>Rahaf Mohammed al-Qunun’s case drew global attention to Saudi Arabia’s strict social rules...”</i>	NB	B ×	NB ✓	NB ✓
<i>‘None of Biden’s Democrat counterparts challenged his claim of economic selflessness.’</i>	B	B ✓	NB ×	B ✓

Table 1.2: Qualitative prediction examples. NB = Not Biased, B = Biased. ✓ = correct, × = incorrect.

Chapter 2

Ece Doga Gül — Sexism Detection

2.1 Methodology

The problem of this work is to detect sexism from social media comments. In this work, the problem of sexism detection is formulated as a binary text classification problem in which a classifier predicts if a comment is sexist or non-sexist. This problem is hard in NLP as sexism is generally expressed implicitly by indirect words and sarcasm.

In this work, three models are experimented:

Bidirectional LSTM. LSTM (Long Short-Term Memory) networks are a specific type of recurrent neural networks capable of learning long term dependencies in sequences [3]. Bidirectional LSTM passes the input sequence forward and backward, so the context is taken into account to a full extent. The network is composed of embedding layer, two stacked bidirectional LSTM layer (with 256 hidden dimension), and a linear classification layer taking average of final hidden states.

BERT. BERT is a pre-trained language model using the Transformer encoder architecture [5]. The model is pre-trained on a massive corpus of text using masked language modeling and next sentence prediction tasks, thereby learning deep contextualized representations. A linear classification layer is added over the [CLS] token’s representation. We use the `bert-base-uncased` version.

RoBERTa. RoBERTa is a version of BERT where the next sentence pre-

diction objective is removed, it uses dynamic masking, and it is pre-trained with bigger batches and more data. [6]. Performance gains on most tasks are usually observed with this change. We are using roberta-base.

Fine tuning for the transformer models such as BERT and RoBERTa. The LSTM is trained from scratch. AdamW optimizer and Adam optimizer are used accordingly.

2.2 Experiments & Findings

Dataset. The EDOS (Explainable Detection of Online Sexism) dataset from SemEval-2023 has been used for this project. [?]. In the EDOS, there are 20,000 English social media comments from Reddit and Gab with labels of sexism or not. It’s been split into 14,000 training, 2,000 development, and 4,000 test examples already. One thing to consider about this dataset is the class imbalance since it is estimated that only 24.3% are sexist while 75.7% are not.

Training Configuration. We trained all models for 3 epochs. The transformer models used the AdamW optimizer with a learning rate of 2×10^{-5} and a batch size of 32. Every input sequence was padded or truncated to 128 tokens. I trained the LSTM using Adam as the optimizer, with a learning rate of 1×10^{-3} . The embedding layer had a size of 128. It has two bidirectional layers, each with a hidden size of 256. All my experiments ran on an NVIDIA GeForce RTX 5070 Laptop GPU.

Results. Table 2.1 represents the performance of each model in the EDOS test set. The metrics are accuracy and macro F1-score.

Model	Accuracy	Macro F1
LSTM	0.8365	0.7530
BERT	0.8565	0.8115
RoBERTa	0.8545	0.7981

Table 2.1: Model performance for sexism detection on the EDOS dataset.

Figure 2.1 shows the accuracy and macro F1 of all three models.

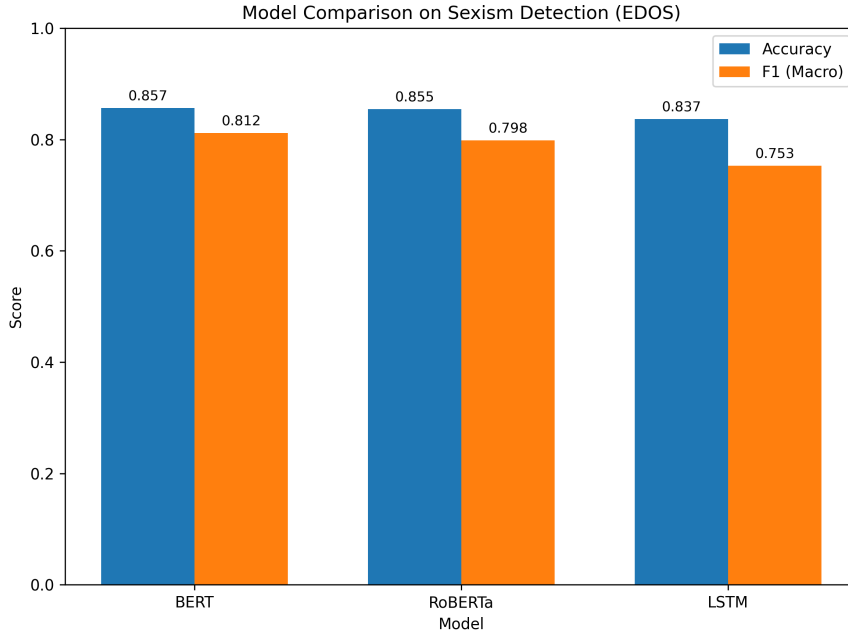


Figure 2.1: BERT achieves best performance. Both transformer models outperforms the BiLSTM.

2.3 Discussion

It is noteworthy the difference between the performance of the transformer pre-trained models and the LSTM. With an accuracy of 83.65% and a macro F1 of 0.7530, the Bidirectional LSTM appears to have similar accuracy compared to transformer models, but the macro F1 is much lower, indicating its poor performance at correctly predicting the minority (sexist) class, which is to be expected given the class imbalance and lack of pre-training.

With 85.65% and 0.8115 macro F1, BERT outperformed all other models, this can be considered proof for transfer learning, the transformer was able to encode context based representation that was able to capture the inherent sexism that the LSTM failed to pick up.

Interestingly, RoBERTa (85.45% accuracy and a macro F1 of 0.7981) scored lower than BERT, despite RoBERTa generally outperforming BERT on downstream tasks. It is possible that, due to the small number of training epochs and the moderate dataset size, RoBERTa was unable to make full use of its potential. The gap is slight, however, and the superiority of the transformer models over the LSTM remains clear.

The macro F1 scores remain lower than the accuracy across all models because the equal weighting of both classes in macro F1 penalizes models that perform poorly on the minority (sexist) class. This makes macro F1 a better indication of how well the model actually performs on the task where the classes are imbalanced.

2.4 Conclusion

Three architectures are evaluated for the sexism detection. In light of the experimental results it is clear that transformer based architectures pre-trained and then fine-tuned are performing better than the baseline LSTM architecture and among transformer architectures, BERT achieved highest accuracy of 85.65% and Macro F1 of 0.8115. These results demonstrate the benefit of pretraining for identifying implicit and context-dependent sexist language, which is difficult to learn from scratch on imbalanced data.

Chapter 3

[Student 3 Name] — [Student 3 Topic]

3.1 Methodology

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3.2 Experiments & Findings

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3.3 Discussion

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3.4 Conclusion

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